

EXAM Linear Control Design 1

31 May 2021 Questions - Multiple Choice

1. Q1. What is the transfer function of this linear system in Fig.1?

(1 Point)

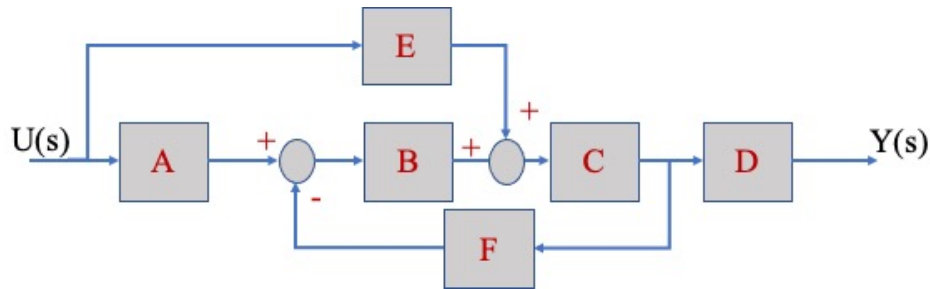


Figure 1: Block Diagram

1. $\frac{Y}{U} = \frac{ABD + DEF^2}{1 + BCF}$
2. $\frac{Y}{U} = \frac{ABCD + ECD}{1 + BCF}$
3. $\frac{Y}{U} = \frac{ABCD + EF}{BCF}$
4. $\frac{Y}{U} = \frac{ABCD + ECD + EC^2FBD}{1 + CDE}$
5. $\frac{Y}{U} = \frac{ABCD}{1 + BCF} + ECD$

CORRECT

Answer:

Q1. [CORRECT n.2]

2. Q2. Consider an *RLC* circuit in Laplace domain as the one in Fig. 2.

What is the corresponding block diagram?

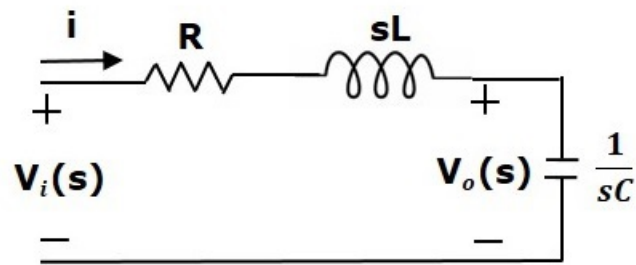


Figure 2: RLC circuit

1. CORRECT
- 2.
- 3.
- 4.
- 5.

Answer:

Q2. [CORRECT n.1]

3. Q3. A system has poles and zeros as shown in Fig. 3 (the circle marks the zero, the cross marks the pole). **Which is the corresponding Bode plot?** (1 Point)

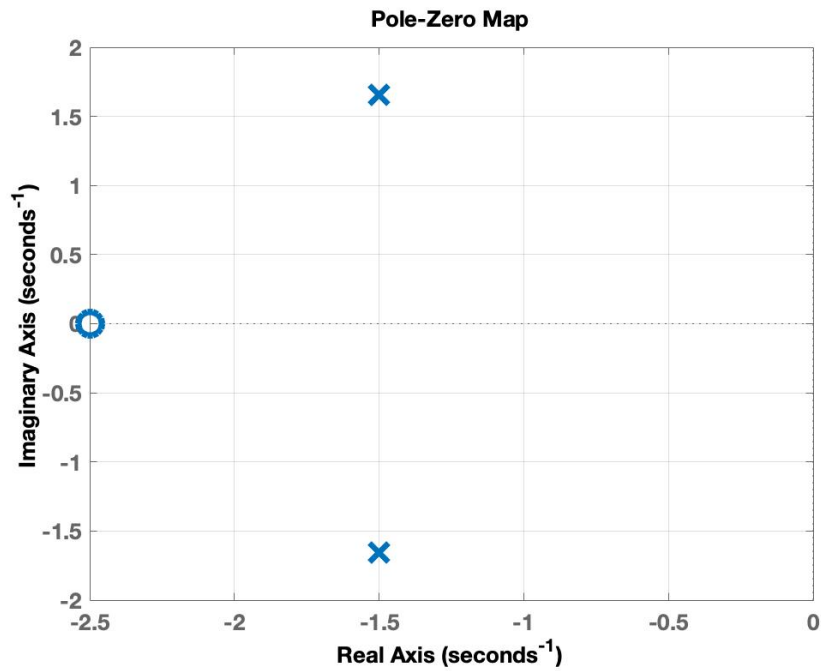
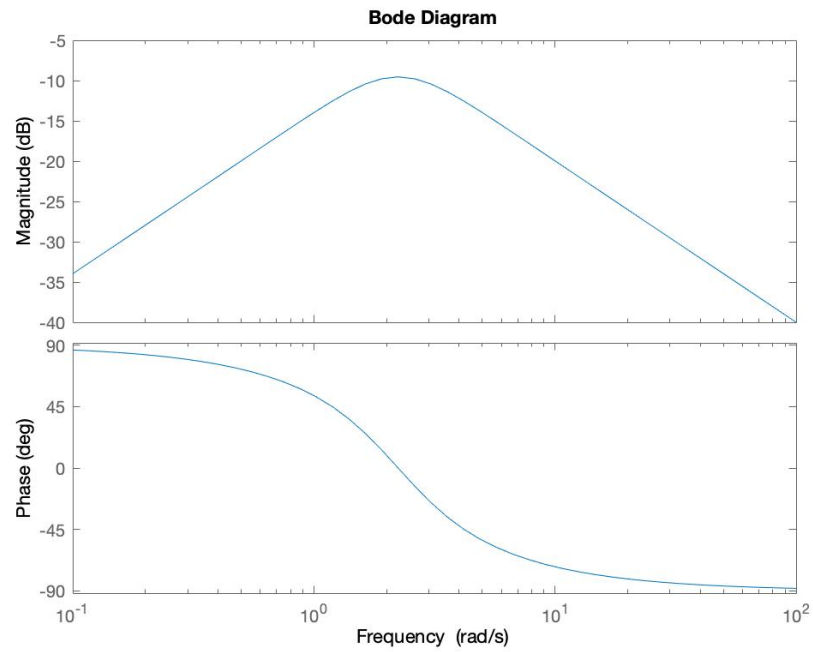
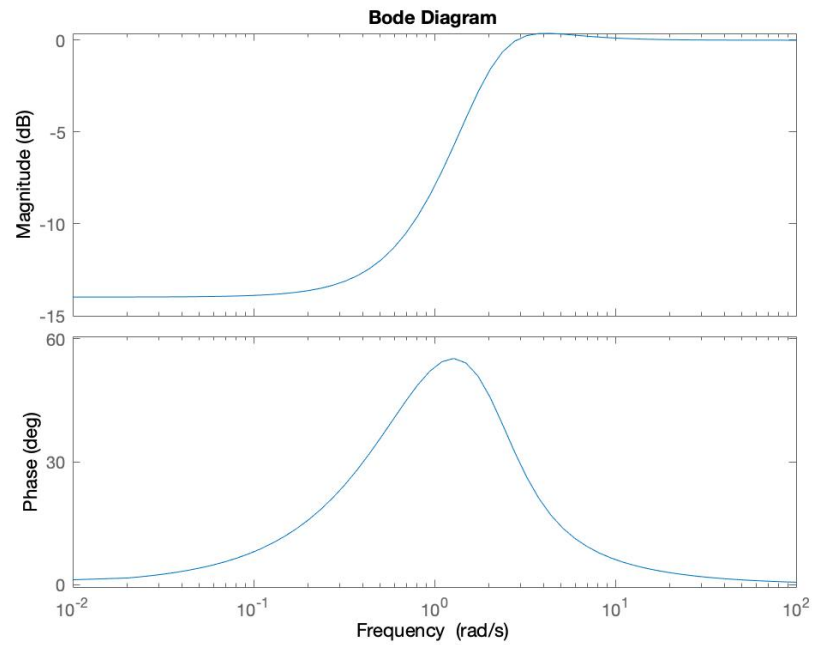


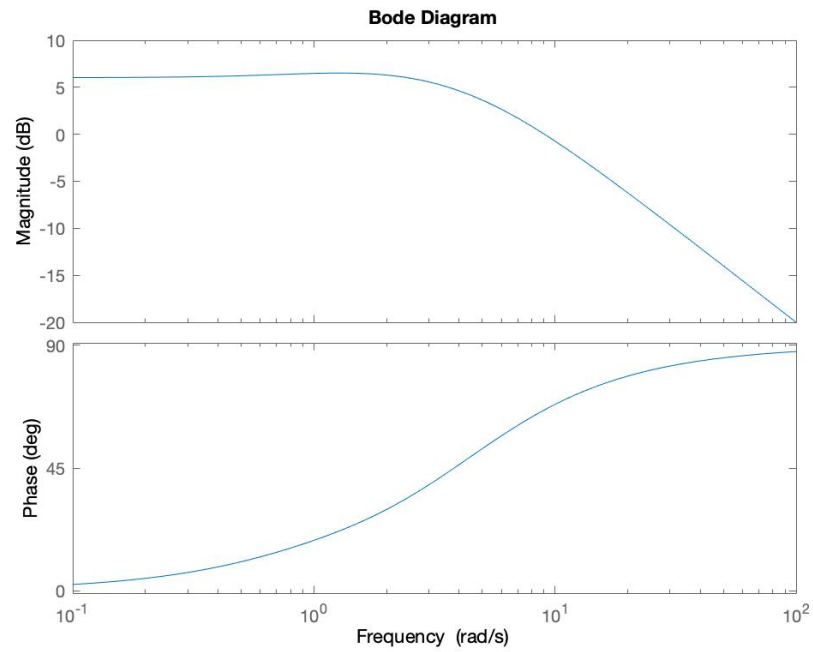
Figure 3: Poles/zeros



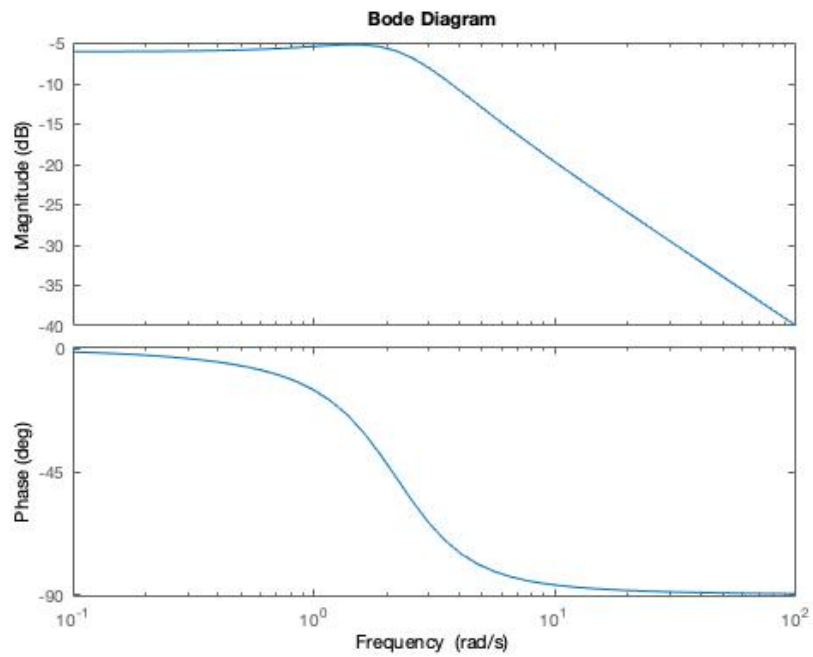
1. (zero in the origin, negative complex poles -1.5)



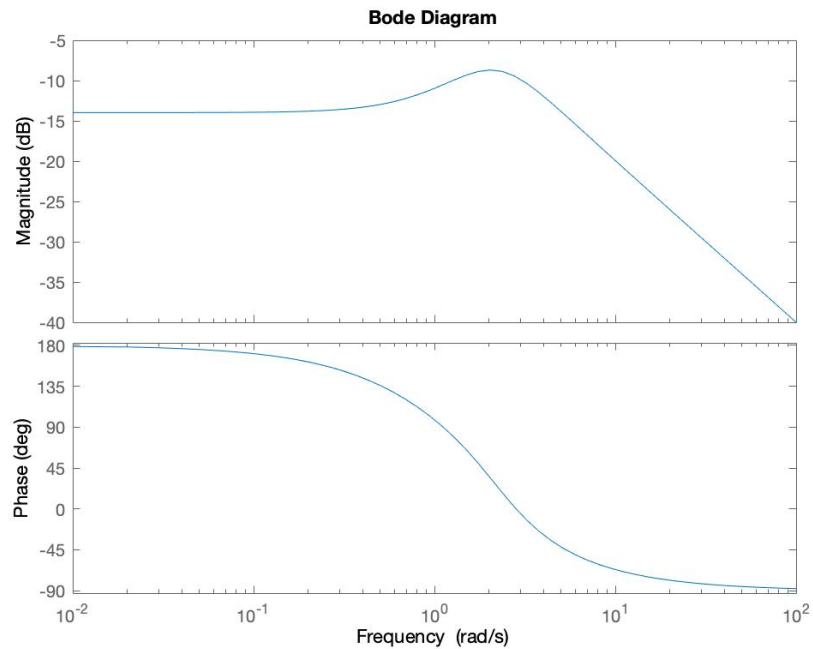
2. (double zero -1, negative complex poles -1.5)



3.
(zero -1, positive and negative real poles 4.19, -1.19)



4.
(CORRECT - negative zero -2.5, negative complex poles - 1.5)



5. (positive)
zero +1, two complex poles (-1.5)

Answer:

Q3. [CORRECT n.4]

4. Q4. The following closed-loop system is given in Fig. 4. Where

$$P(s) = \frac{1}{(s+1)^3}$$

is the system transfer function and K is a proportional controller with gain $K \in \mathbb{R}^+$.

For which values of K is the closed loop stable? (1 Point)



Figure 4: Closed-loop system

1. $K \geq 8$

2. $-1 < K < 6$
3. $0 < K < 8$ correct
4. $K > -10$

Answer:

Q4. **[CORRECT n.3]** To find the K values for which the system is asymptotically stable we can use the Bode criterion. The system has the transfer function $L(s) = K/(s+1)^3$ and it satisfies the conditions of Bode criterion because: 1) 3 poles $p_1=p_2=p_3 = -1 < 0$ negative real part so it is stable; 2) the Bode plot of the magnitude depends on K. For $K = 10$, it has a monotonous decreasing trend. It is sufficient that the gain is greater than 1 in module for the diagram to start from a value greater than 0 dB and the crossover frequency is well defined. Since the *gain* = K, the crossover frequency is well defined for any value $K > 1$. To verify the asymptotic stability of the closed loop we have to consider if the *gain* > 0 and *PM* > 0. *gain* > 0: as regards the gain, since the gain of the transfer function = K we can say that the first condition of the Bode criterion is verified for every $K > 1$. *PM* > 0: the phase diagram is independent of K ($K > 0$). The phase margin can be identified from the diagram. As K increases, the crossover frequency increases and therefore the modulus diagram shifts up and PM decreases. Since for Bode, the closed loop system is asymptotically stable if and only if *PM* > 0, we want to understand what is the limit value of K beyond which *PM* ≤ 0. It corresponds to having a critical phase -180° or *PM* = 0° . To find the limit value of K one must:

1. find ω such that the phase of $L(s) = -180^\circ$.

$$\angle L(s) = \angle K - 3\angle(1+j\omega) = 0^\circ - 3\arctan \omega = -3\arctan \omega \quad (1)$$

then solve the equation:

$$-3\arctan \bar{\omega} = -180^\circ \quad \arctan \bar{\omega} = 60^\circ \quad \bar{\omega} = \tan 60^\circ = \sqrt{3} \quad (2)$$

2. Calculate $\bar{K}$ so $|L(j\bar{\omega})| = 1$

$$|L(j\bar{\omega})| = |L(j\sqrt{3})| = \left| \frac{K}{(1+j\sqrt{3})^3} \right| = \frac{K}{(\sqrt{1+3})^3} = \frac{K}{8} \frac{\bar{K}}{8} = 1 \rightarrow [\bar{K} = 8] \quad (3)$$

therefore, the closed-loop system is asymptotically stable if and only if: $K < \bar{K} = 8$ If $K \geq 8$ the closed-loop system is not stable. It is possible to analyze the response to the step as K varies up to such a limit.

5. Q5. The bode plot diagrams of the magnitude and phase of a transfer function G(s)

of a linear system is shown in Fig. 5.

Which of the following answers is correct?

(1 Point)

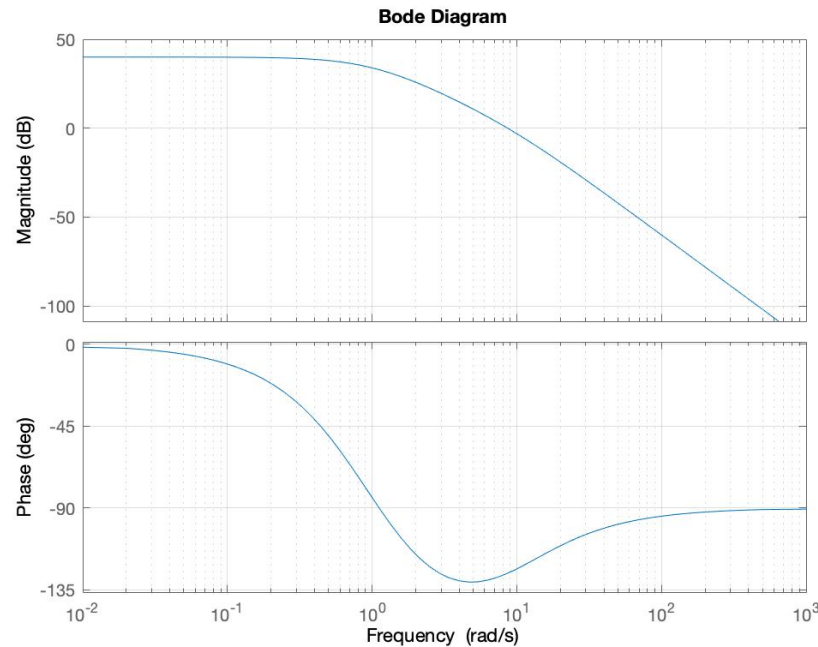


Figure 5: Bode Plot Diagrams - Q5

1. $G(s)$ has negative and positive real poles and no zeros
CORRECT
2. $G(s)$ has only 1 negative real pole and 1 negative real zero
Looking at the Bode diagram, there is more than one pole
3. $G(s)$ has only 1 positive real pole and no zeros
The magnitude would end in 0 dB
4. $G(s)$ has positive real poles and 1 negative real zero
The phase would start at -180 and increase

Answer:

Q5. [**CORRECT n.1**] DC gain 40 dB equal to 100 (magnitude horizontal and initial phase is 0 degrees.. Magnitude decreases from 0 dB/decade to -40 dB/decade at $\omega = 1$. Correspondingly, the phase changes 180 degrees. This means that there are two poles with abs 1. There are no peaks, so poles are reals. moreover, the phase in $\omega = 1$ decreases of 180 degrees so both poles are

negatives. At $\omega = 10$ the slope of the magnitude decreases from -40 dB/decade to -60 dB/decade and the asymptotic phase increases of 90 degrees. This concludes that $G(s)$ has one pole +10. In conclusion, DC gain =10, poles $p_1=p_2=-1$ and $p_3=10$, and no zeros.

$$G(s) = \frac{100}{(1+s)^2 * (1-0.1*s)}$$

6. Q6. A closed-loop system has a loop transfer function

$$L(s) = \frac{K}{s(s+3)(s+10)}$$

and the following Bode plot in Fig.6

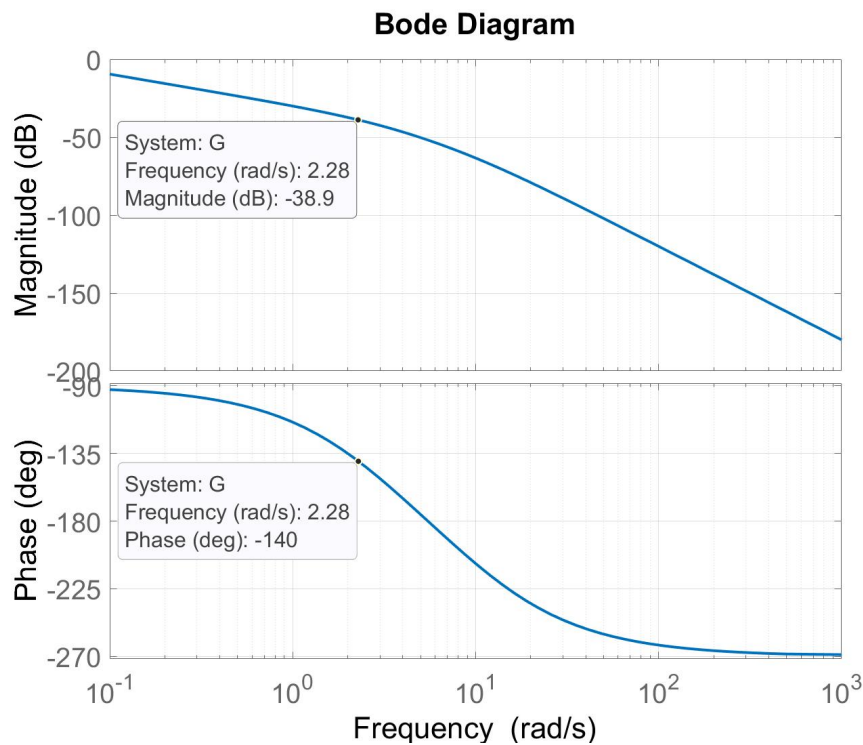


Figure 6: Bode Plot - Q6

What is the gain K so that the phase margin is $PM=40^\circ$?
(1 Point)

1. $K = 19.5$ Wrong, this is $38.9/2$
2. $K = 90$ Wrong, $PM=39.3$, $K = 90$
3. $K = 44$ Wrong, this is $88/2$
4. $K = 38.9$ Wrong, this is $K=88$ in dB
5. $K = 88$ CORRECT, $db2mag(38.9) = 88.1$

Answer:

Q6. [CORRECT n.5]

7. Q7. Which is the transfer function $\frac{X_2(s)}{F(s)}$ for the system shown in Figure 7? Both masses slide on a frictionless surface and $k = 1$ N/m. (Consider zero initial conditions). [hint = write the equations of motion]

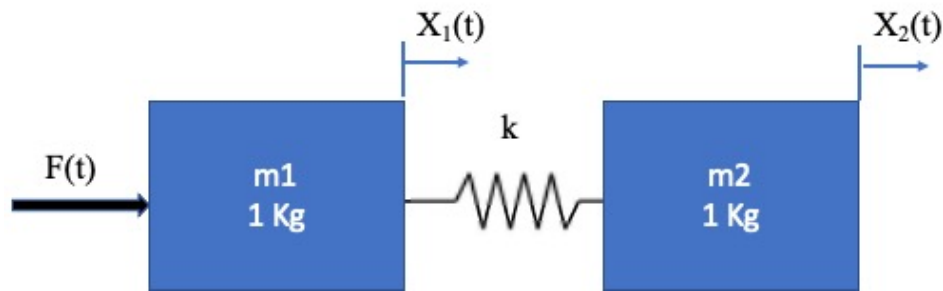


Figure 7: Two connected masses on a frictionless surface

(1 Point)

1. $\frac{X_2(s)}{F(s)} = \frac{1}{(s^2) * (s^2 + 1)}$
2. $\frac{X_2(s)}{F(s)} = \frac{1}{s^2 * (s^2 + 2)}$
3. $\frac{X_2(s)}{F(s)} = \frac{1}{s^2 * (s + 2)}$
4. $\frac{X_2(s)}{F(s)} = \frac{1}{s * (s^2 + 1)}$

CORRECT

5.

$$\frac{X_2(s)}{F(s)} = \frac{1}{(s^4 + 1)}$$

Answer:

Q7. [CORRECT n. 2] Equations of motion:

$$m_1 \ddot{x}_1 + k(x_1 - x_2) = F \qquad m_2 \ddot{x}_2 + k(x_2 - x_1) = 0$$

Taking the Laplace transform (with zero initial conditions) and solving for $X_2(s)$ yields:

$$X_2(s) = \frac{k}{(m_2 * s^2 + k)(m_1 * s^2 + k) - k^2} F(s)$$

Then, with $m_1 = m_2 = 1$

$$\frac{X_2(s)}{F(s)} = \frac{1}{s^2 * (s^2 + 2)}$$

8. Q8. Consider the differential equation

$$\ddot{y}(t) + 2\dot{y}(t) + y(t) = u(t)$$

Where $y(0) = \dot{y}(0) = 0$ and $u(t)$ is the input.

Which are the poles of this system?

(1 Point)

1. $s_1 = 2, s_2 = -1$
2. $s_1 = -1, s_2 = -1$ **CORRECT**
3. $s_1 = 1j, s_2 = -1j$
4. $s_1 = -1, s_2 = -2$
5. $s_1 = -j, s_2 = -j$

Answer:

Q8. [CORRECT n. 2]

$y(0) = \dot{y}(0) = 0$ (No need to linearize a second order linear system)

Laplace transform

$$s^2 * Y(s) + 2s * Y(s) + Y(s) = U(s)$$

Transfer function

$$\frac{Y(s)}{U(s)} = \frac{1}{s^2 + 2s + 1}$$

and the poles are: $s_1 = -1, s_2 = -1$

9. Q9. For the unit feedback system shown in Fig.8, Which is the gain K of the proportional controller so that the output $y(t)$ has an overshoot of no more than 12% in response to a unit step?



Figure 8: Unity feedback system for Q8.

(1 Point)

1. $0 \leq K \leq 20$ **CORRECT**
2. $5 \leq K \leq 25$
3. $K \geq 25$
4. $-1.5 \leq K \leq 10$

Answer:

Q9. [**CORRECT** n. 1]

$$\frac{Y(s)}{R(s)} = \frac{K}{s^2 + 5s + K} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad \omega_n = \sqrt{K}$$

$$\zeta = \frac{5}{2\omega_n} = \frac{2.5}{\sqrt{K}} \quad (4)$$

In order to have an overshoot no more than 12%:

$$M_p = e^{-\zeta \omega_n / \sqrt{1-\zeta^2}} \leq 0.12$$

$$\text{Solving for } \zeta: \zeta = \sqrt{\frac{(\ln M_p)^2}{\pi^2 + (\ln M_p)^2}} \geq 0.559$$

Using (4) and the solution for ζ :

$$K = \frac{6.25}{\zeta^2} \leq 19.97$$

$$0 \leq K \leq 20$$

10. Q10. A certain system is represented by the Bode diagram (only the magnitude in dB) shown in Fig. 9 (no zeros in the RHP). **Which is the correct answer showing the response of this system to a unit step input (assuming zero initial conditions)?**

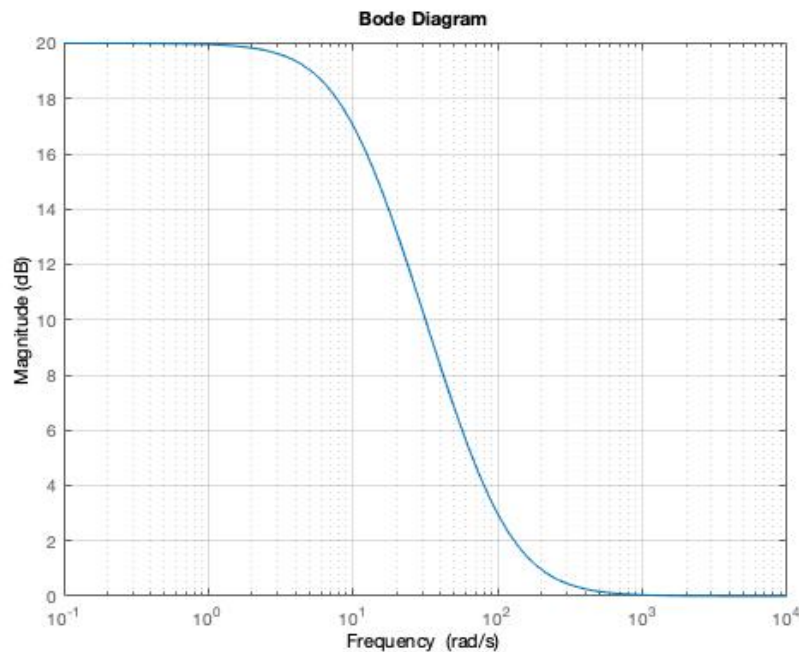
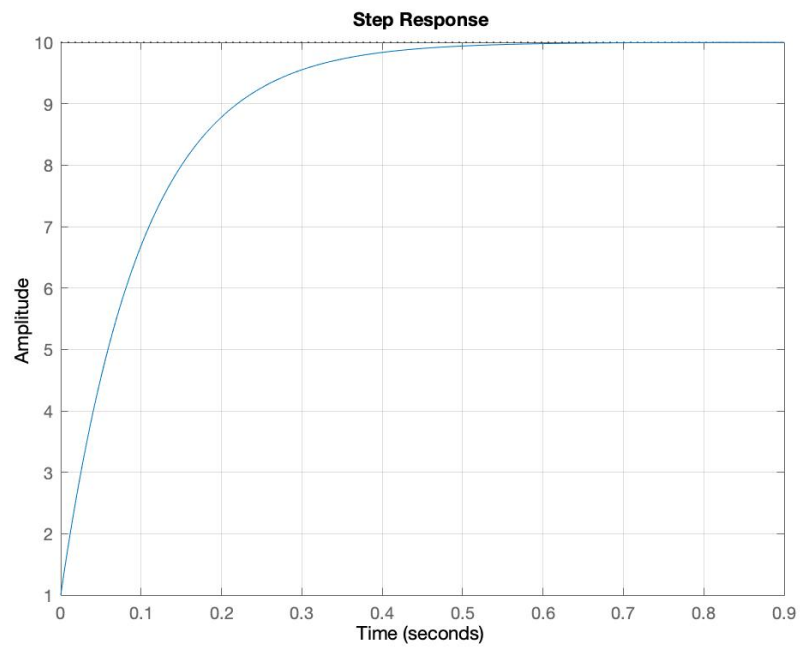
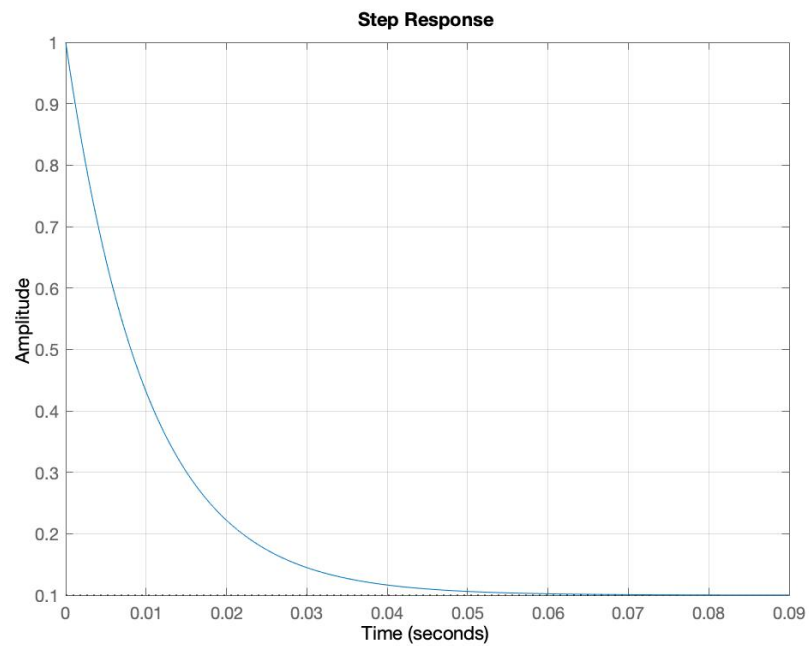


Figure 9: Magnitude portion (dB) of the Bode diagram for Q10.

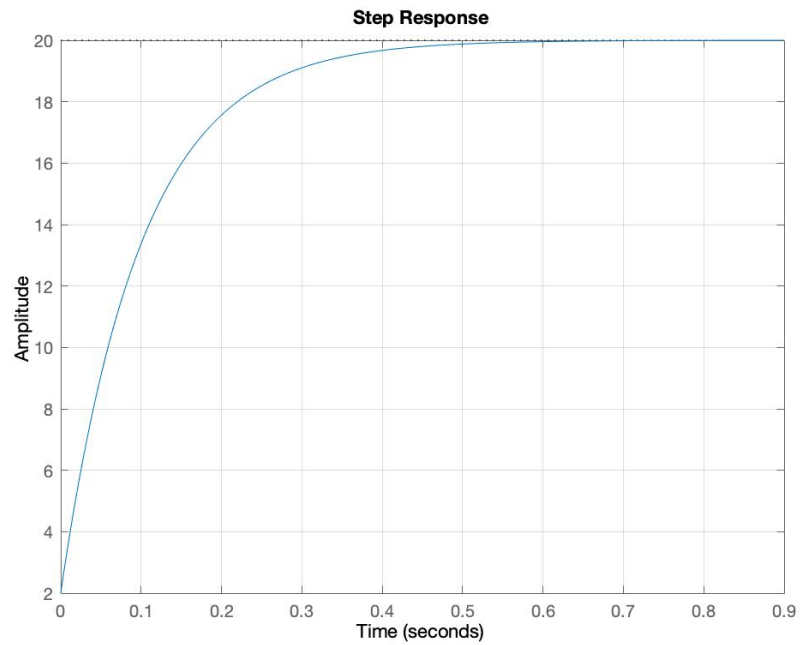
(1 Point)



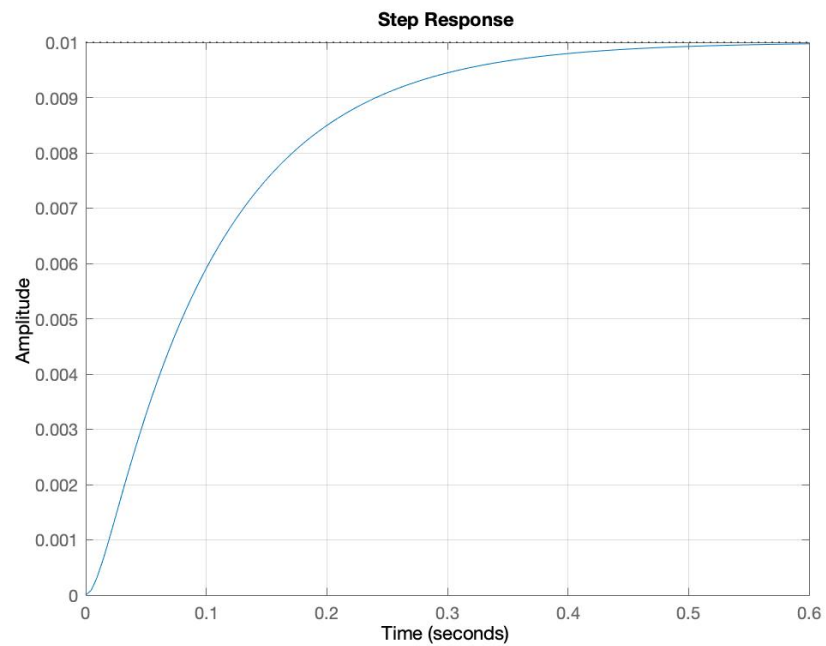
1.
CORRECT



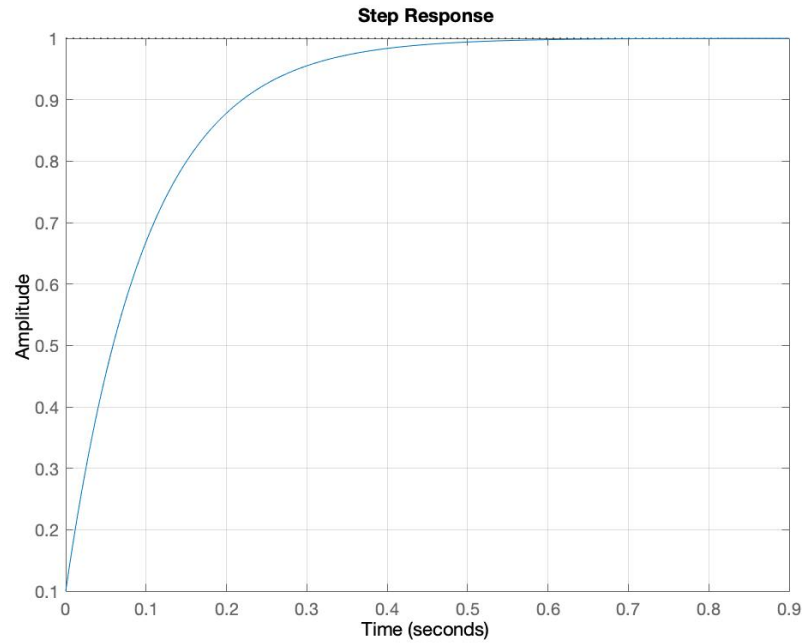
2.
inverted num. and den.



3.
used gain in dB - 20



4.
two poles $s = -10$ and $s = -100$, no zero



5.

without the gain 10

Answer:

Q10. [CORRECT n. 1]

Through the inspection of the given Bode plot, the transfer function is the following:

$$G(s) = \frac{10 * (s/100 + 1)}{(s/10 + 1)} = \frac{s + 100}{(s + 10)}$$

The response to a unit step input is $Y(s) = G(s)U(s)$

$$Y(s) = \frac{s + 100}{(s + 10)s} = \frac{10}{s} - \frac{9}{(s + 10)} \quad y(t) = \mathcal{L}^{-1}Y(s) = 10 \times 1(t) - 9e^{-10t}$$

Question 11

Consider a system with transfer function

$$G(s) = \frac{120}{s^3 + 43s^2 + 120s}$$

and Bode plot as shown in the following figure. A P-controller K_P is applied to the system

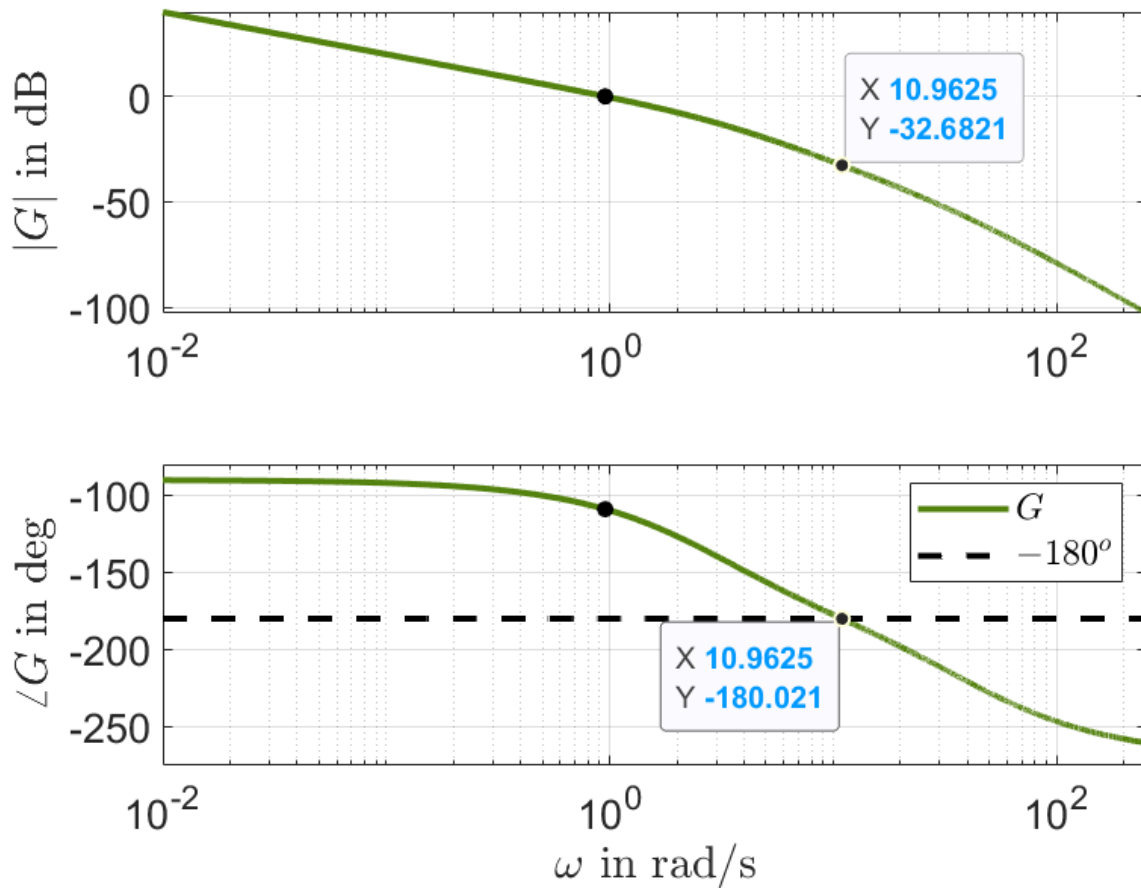


Figure 1: Bode plot

and the loop is closed with unit feedback. The closed-loop system is stable for:

- a) $K_P = -1$ (Negative gain)
- b) $K_P = 25$ (Correct)
- c) $K_P = 71$ (Exceeds gain margin $K_M = 32.6821$ dB = 43)
- d) $K_P = -32$ (Negative gain)
- e) $K_P = 45$ (Exceeds gain margin $K_M = 32.6821$ dB = 43)

Question 12

Consider the system described by the following differential equation

$$y^{(4)}(t) + 9y^{(3)}(t) + 20\ddot{y}(t) = 71u(t)$$

where $\ddot{y}(t) = \frac{d^2y}{dt^2}$ and $y^{(n)}(t) = \frac{d^ny}{dt^n}$ denotes the n^{th} time derivative of $y(t)$. Which of the following statements is the correct one?

- a) The system is stable. (Wrong)
- b) The system is nonlinear. (It is linear)
- c) The system has two zeros. (It has no zeros)
- d) The system is unstable. (Correct)
- e) The steady state (DC) gain of G is 3.55. (It is infinite)

Solution: The system is written in the Laplace domain as

$$s^4y(s) + 9s^3y(s) + 20s^2y(s) = 71u(s) \Rightarrow G(s) \triangleq \frac{y(s)}{u(s)} = \frac{71}{s^4 + 9s^3 + 20s^2} = \frac{71}{s^2(s^2 + 9s + 20)}$$

The transfer function has a double pole at the origin and the system is therefore unstable.

Question 13

Consider a system with open-loop transfer function

$$G(s) = \frac{s+1}{s^2+43s-90}$$

and Nyquist plot as shown in the following figure. The Nyquist plot intersects the real axis along the counter-clockwise direction at the point $(-0.0111, 0)$. A P-controller K_P is applied to the system and the loop is closed with unit feedback. The closed-loop system is stable for:

- a) $K_P = 100$ (Correct)
- b) $0.02 < K_P < 1$ (Too small gain)
- c) $0 < K_P < 80$ (Too small gain)
- d) $K_P < -0.0111$ (Negative gain)
- e) $0 < K_P < 0.02$ (Too small gain)

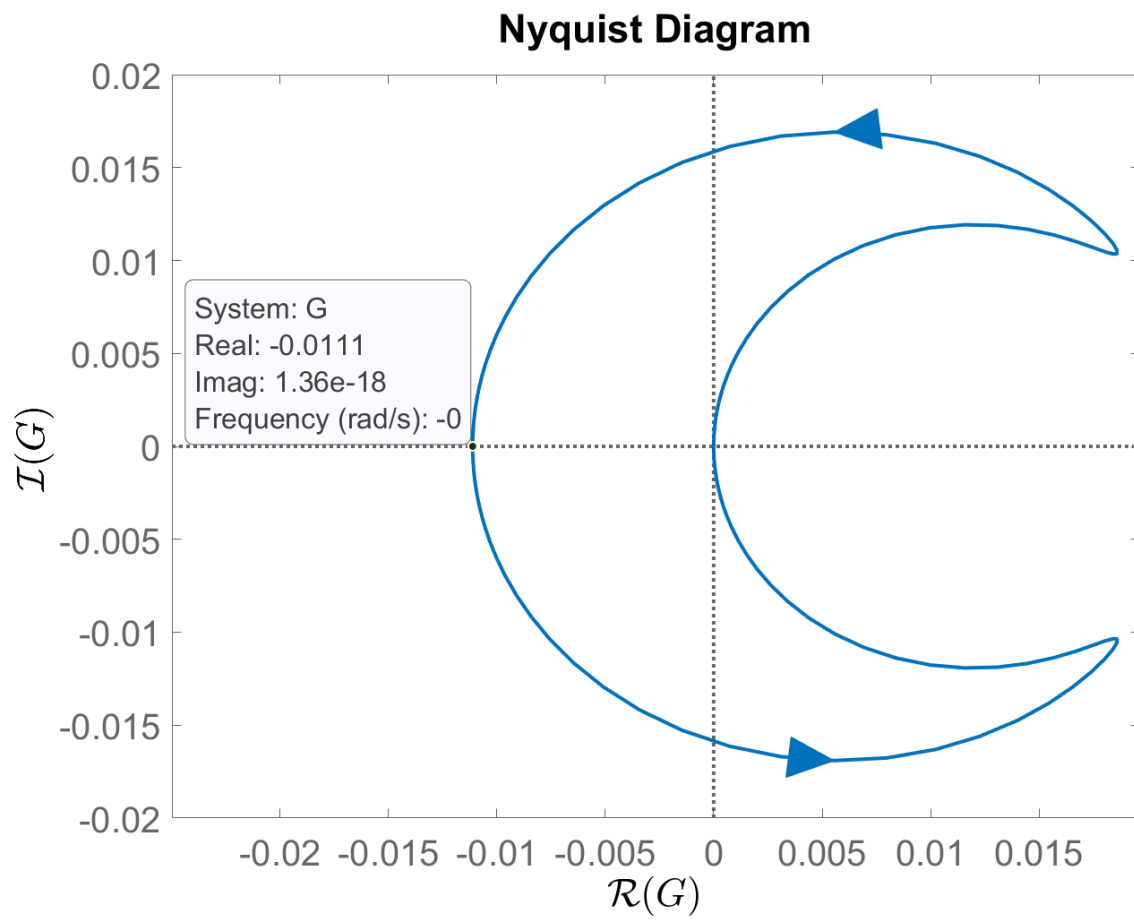


Figure 2: Nyquist plot

Solution: The system is unstable with one pole in the RHP. We need to ensure that the Nyquist curve encircles the point $(-1,0)$ once along the CCW direction. Already the direction is ok, so we only need to adjust the gain K_P . The intersection point of the curve with the negative real axis is at -0.0111 , so multiplying G by $K_P = \frac{1}{0.0111} \approx 90$ (“gain margin” if we did not have unstable poles), we will make the curve cross exactly the critical point $(-1,0)$. Hence a proportional gain $K_P > 90$, will ensure one encirclement of the critical point, which is needed for closed-loop stability. Negative gain will simply flip the curve about the imaginary axis. Therefore, the only acceptable option is (a).

Question 14

Consider a system with transfer function

$$G(s) = \frac{1200}{s^3 + 54s^2 + 570s + 400}$$

and Nyquist plot (only positive frequencies) as shown in the following figure. The Nyquist

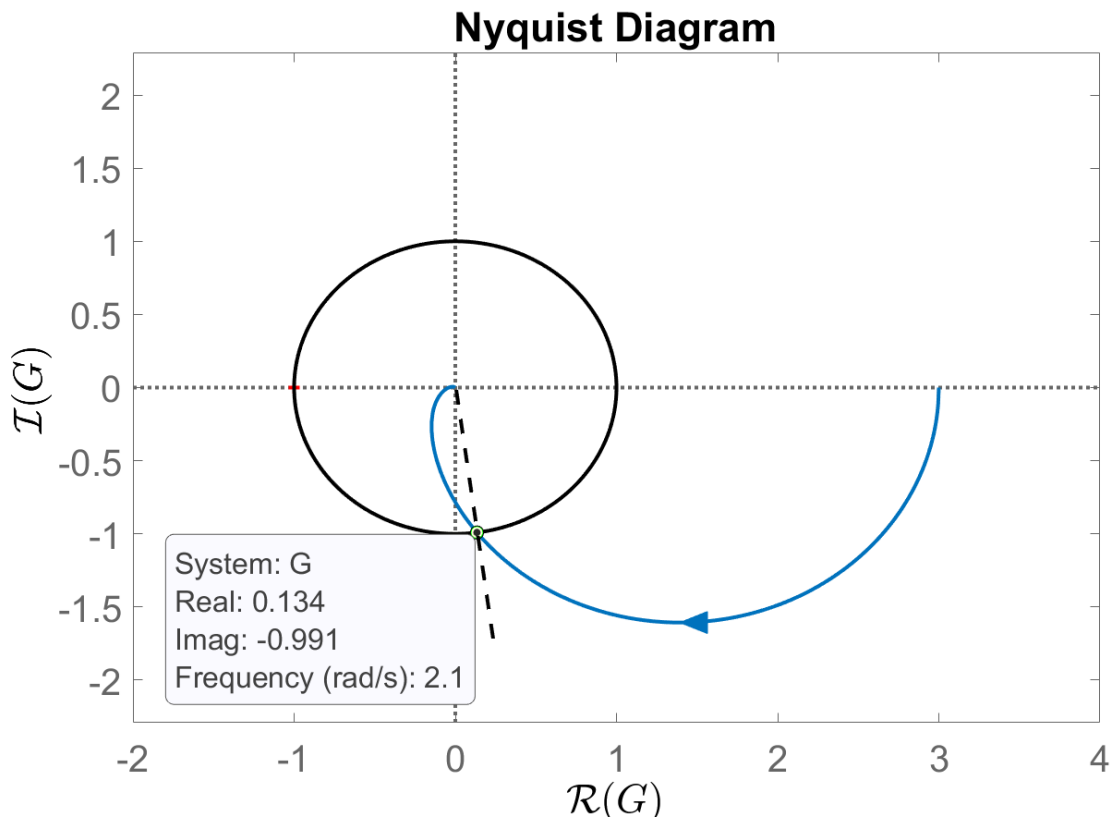


Figure 3: Nyquist plot

plot intersects the unit circle along the clockwise direction at the point $(0.134, -0.99)$. The phase margin of the system is approximately

- a) $\gamma_M = 0.1354^\circ$ (Wrong)
- b) $\gamma_M = 1.705^\circ$ (In rad)
- c) $\gamma_M = -1.44^\circ$ (ϕ_c in rad)
- d) $\gamma_M = 82.29^\circ$ ($-\phi_c$)
- e) $\gamma_M = 97.7^\circ$ (Correct)

Solution: The phase margin is defined as $\gamma_M = 180^\circ + \phi_c$, where ϕ_c is the phase of the transfer function at the crossover frequency. This angle is confined between the positive real axis (0°) and the dashed line in the Nyquist diagram, which starts from the origin and crosses the point of intersection of the Nyquist curve with the unit circle (0 dB magnitude). It is calculated as $\phi_c = \arctan\left(\frac{-0.99}{0.134}\right) \cong -82.29^\circ$. The phase margin is then found by $\gamma_M \cong 180^\circ - 82.29^\circ = 97.7^\circ$. Therefore, the only acceptable option is (e).

Question 15

Consider a second-order type-0 system with two stable real poles and no zeros. A P-controller with $K_P > 1$ is applied to the system. Which of the following statements is the correct one?

- a) Increasing K_P will shift the phase curve in the open-loop Bode plot upwards. (K_P does not affect the phase curve in the Bode plot)
- b) Increasing K_P will increase the crossover-frequency of the open-loop system. (Correct)
- c) Increasing K_P too much can theoretically make the closed-loop system unstable. (An open-loop stable second-order type-0 system with no zeros has infinite gain margin)
- d) After a finite value of K_P (i.e. $K_P < \infty$), the steady state error in the output of the closed-loop system due to a step change in the reference signal will be zero. (Type-0 system cannot have zero steady state error if the loop is closed with a finite proportional gain)
- e) The open-loop system can have multiple crossover-frequencies. (The system has real poles and therefore no resonance peaks that could cause the phenomenon of multiple crossings of the 0 dB line)

Question 16

Consider a system with transfer function

$$G(s) = \frac{1224}{s^3 + 30s^2 + 257s + 612}$$

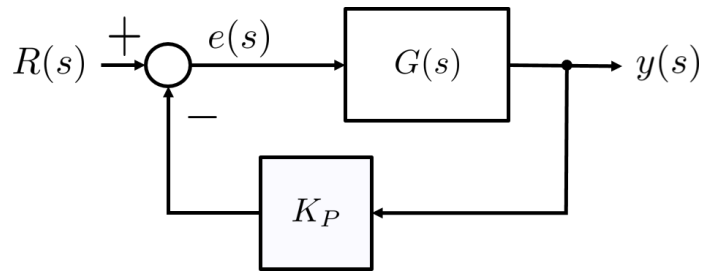


Figure 4: Closed-loop system.

A P-controller with $K_P = 2$ is applied to the system but it is placed in the feedback branch as in the following figure. The steady state error $e(s)$ in the closed-loop system due to a unit step change in the reference signal is:

- a) $e_{SS} = 0$ (Type-0 system)
- b) $e_{SS} = 2$ (DC gain of $G(s)$)
- c) $e_{SS} = 0.2$ (Correct)
- d) $e_{SS} = 0.4$ (DC gain of closed-loop transfer function)
- e) $e_{SS} = 0.8$ (DC gain of $\frac{K_P G(s)}{1 + K_P G(s)}$)

Solution: The transfer function from the reference signal to the error $e(s)$ is given by

$$G_{er}(s) = \frac{1}{1 + K_P G(s)} = \frac{s^3 + 30s^2 + 257s + 612}{s^3 + 30s^2 + 257s + 3060} .$$

The steady-state error is found by

$$e_{ss} = \lim_{s \rightarrow 0} e(s) = \lim_{s \rightarrow 0} s G_{er}(s) \frac{1}{s} = 0.2 .$$

Therefore, the only acceptable option is (c).

Question 17

Consider the system described by the following transfer function and Bode plot

$$G(s) = \frac{5(s+7)(s+12)}{(s+4)(s+9)(s+17)(s+5)(s+70)}$$

A P-Lead-Lag controller

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}}$$

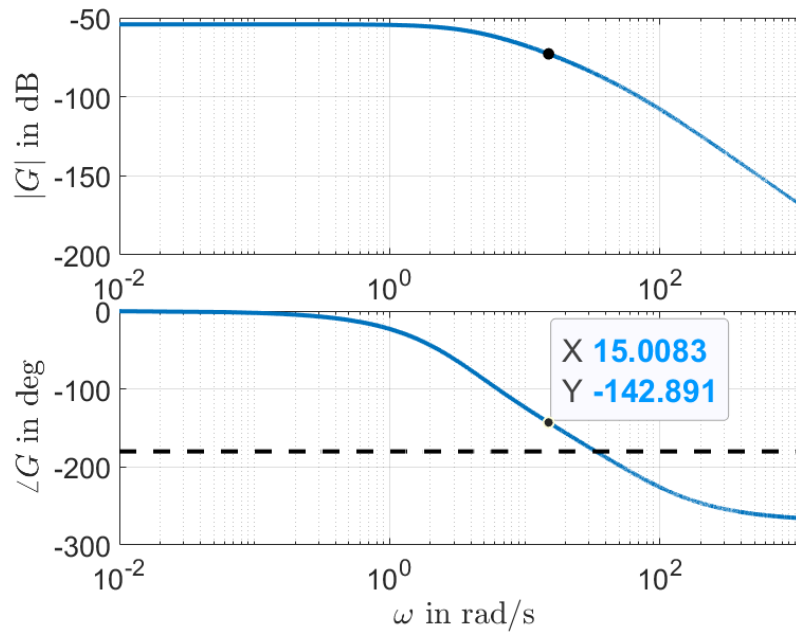


Figure 5: Bode plot

with $\alpha = 0.2$ needs to be designed for the system. The new crossover frequency (after applying the controller) is selected as $\omega_c = 15$ rad/s and the phase contribution of the Lag part at that frequency is given in radians by

$$\phi_L = \arctan \left(N_i \frac{1 - \beta}{1 + \beta N_i^2} \right) ,$$

where $N_i = \omega_c \tau_i = 3$. If a phase margin $\gamma_M = 70^\circ$ is desired for the open-loop system $C(s)G(s)$, what is the value of the parameter β of the Lag-part?

- a) $\beta = 0$ (β must be positive)
- b) $\beta = 2$ (Correct)
- c) $\beta = -1.5$ (β cannot be negative)
- d) $\beta = 0.3$ (Used degrees instead of radians for Lag phase)
- e) $\beta = 0.62$ (Included I-part too)

Solution: Let ϕ_G, ϕ_d, ϕ_L be the phase contributions of G , Lead and Lag parts of the open-loop system at the crossover frequency, respectively. The phase of G is given from the Bode plot (or can easily be calculated in Matlab) as $\phi_G = -142.891^\circ$. At $\omega_c = 15$ rad/s the following

holds:

$$\begin{aligned}
 -180^\circ + \gamma_M &= \phi_G + \phi_d + \phi_L \Leftrightarrow \phi_L = -180^\circ + \gamma_M - \phi_G - \phi_d \Leftrightarrow \\
 \arctan\left(N_i \frac{1-\beta}{1+\beta N_i^2}\right) &= -180^\circ + 70^\circ + 142.891^\circ - \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Leftrightarrow \\
 N_i \frac{1-\beta}{1+\beta N_i^2} &= \tan(-8.9193^\circ) \Leftrightarrow \beta = \frac{N_i - \tan(-8.9193^\circ)}{N_i(1 + N_i \tan(-8.9193^\circ))} \Leftrightarrow \beta = 1.9886 \cong 2
 \end{aligned}$$

Therefore, the only acceptable option is (b).

Question 18

A system with transfer function

$$G(s) = \frac{0.7(s + 0.5)}{(5s + 1)(s^2 + 0.2s + 0.6)(0.01s + 1)}$$

has the following Bode plot

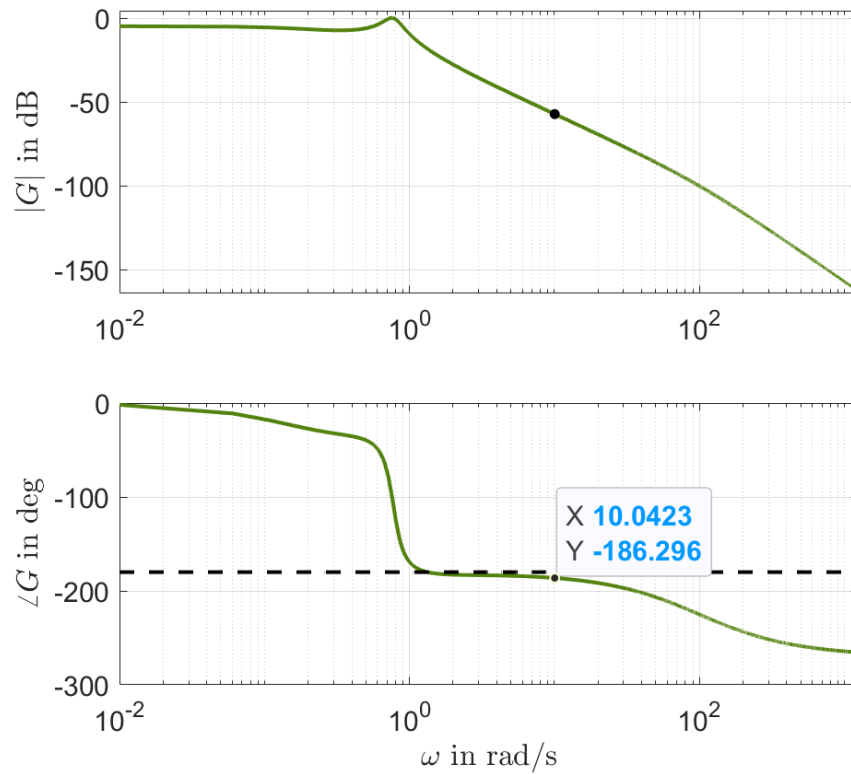


Figure 6: Bode plot

A PI-Lead controller

$$C(s) = K_P \frac{\tau_i s + 1}{\tau_i s} \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 45^\circ$ and crossover frequency $\omega_c = 10$ rad/s. If $N_i = \omega_c \tau_i = 8$, a good selection for α and K_P is:

- a) $\alpha = 0.5$ and $K_P = 500$. (Phase margin too low, 6°)
- b) $\alpha = 2$ and $K_P = 1000$. (Unstable due to negative phase margin)
- c) $\alpha = 0.7$ and $K_P = 591$. (Unstable due to negative phase margin)
- d) $\alpha = 0.08$ and $K_P = 200$. (Correct)
- e) $\alpha = 0.07$ and $K_P = -154$. (Unstable due to negative gain)

Solution: Let ϕ_G, ϕ_d, ϕ_i be the phase contributions of G , Lead and I parts of the open-loop system at the crossover frequency, respectively. The phase of G can easily be calculated in Matlab as $\phi_G = -184.9765^\circ$. At ω_c the following holds:

$$\begin{aligned} -180^\circ + \gamma_M &= \phi_G + \phi_d + \phi_i \Leftrightarrow \phi_d = -180^\circ + \gamma_M - \phi_G + \arctan\left(\frac{1}{N_i}\right) \Leftrightarrow \\ \phi_d &= 58.4213^\circ \Leftrightarrow \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) = 58.4213^\circ \Leftrightarrow \alpha = 0.08 . \end{aligned}$$

From there we can calculate the time constants of the I and Lead parts as $\tau_i = \frac{N_i}{\omega_c} = 0.8$ s and $\tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.3536$ s. With the transfer functions of the controller and the system known, we can calculate the proportional gain K_P such that the open-loop transfer function has zero dB gain at ω_c :

$$K_P = \frac{1}{|G_{ol}(j\omega_c)G_I(j\omega_c)G_{Lead}(j\omega_c)|} = 200.0413 .$$

Therefore, the only acceptable option is (d).

Question 19

Consider the system described by the following block diagram where $G_1(s)$ and $G_2(s)$ are type-0 systems.

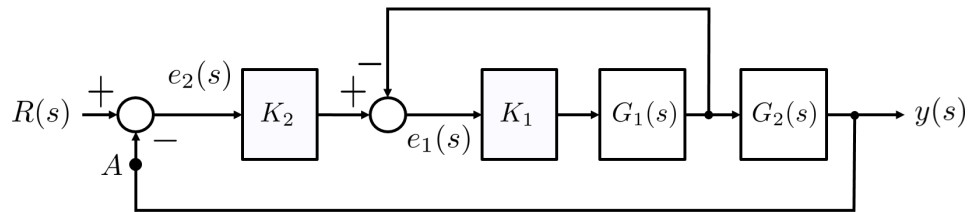


Figure 7: Block diagram.

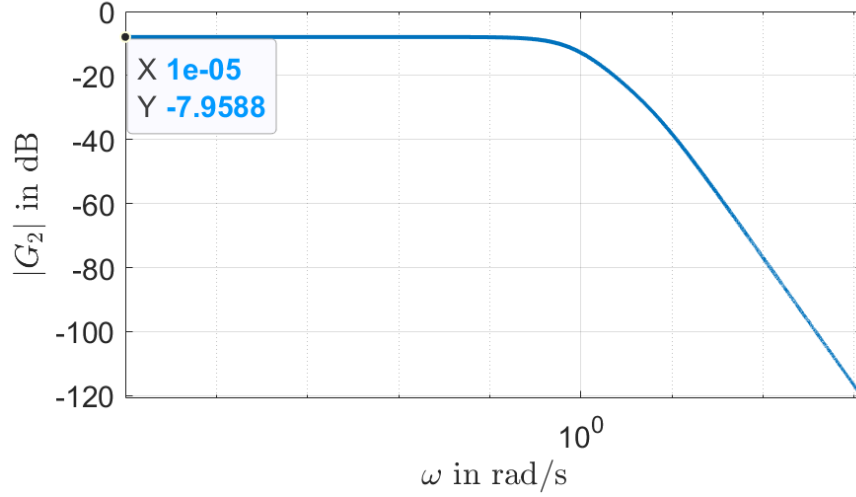


Figure 8: Magnitude Bode plot.

The magnitude Bode plot of $G_2(s)$ is shown below:

When the outer loop is opened at point A , a step change in $R(s)$ equal to $\frac{1}{K_2}$ causes a steady state error $e_1(0) = 0.4$. What should be the value of K_2 , such that a unit step change in the reference $R(s)$, when the loop is closed, causes a steady state error $e_2(0) = 0.05$?

- a) $K_2 = 118.75$. (Used closed-loop system DC gain instead of that from e_2 to e_1)
- b) $K_2 = -3.98$. (Used dB for $G_2(0)$)
- c) $K_2 = -5.97$. (Used closed-loop system DC gain instead of $G_{e_2, e_1}(0)$ and dB for $G_2(0)$)
- d) $K_2 = 91.25$. (Gain margin of entire open-loop)
- e) $K_2 = 79.17$. (Correct)

Solution: Let $\varepsilon_1 \triangleq e_1(0)$ and $\varepsilon_2 \triangleq e_2(0)$. The transfer function from e_2 to e_1 is

$$G_{e_2 e_1}(s) = \frac{K_2}{1 + K_1 G_1(s)}$$

Then

$$\varepsilon_1 = \lim_{s \rightarrow 0} s G_{e_2 e_1}(s) \frac{1}{s K_2} \Leftrightarrow \varepsilon_1 = \frac{1}{1 + K_1 G_1(0)} \Leftrightarrow K_1 G_1(0) = \frac{1 - \varepsilon_1}{\varepsilon_1} \quad (1)$$

Moreover, the transfer function $G_{cl,1}(s)$ from the output of the gain K_2 to the output of $G_1(s)$ is given by

$$G_{cl,1}(s) = \frac{K_1 G_1(s)}{1 + K_1 G_1(s)} \Rightarrow G_{cl,1}(0) = \frac{K_1 G_1(0)}{1 + K_1 G_1(0)} \stackrel{(1)}{=} \frac{\frac{1 - \varepsilon_1}{\varepsilon_1}}{1 + \frac{1 - \varepsilon_1}{\varepsilon_1}} = 1 - \varepsilon_1 \quad (2)$$

The transfer function from the reference $R(s)$ to $e_2(s)$ is

$$G_{e_2R} = \frac{1}{1 + K_2 G_{cl,1}(s) G_2(s)}$$

Then

$$\varepsilon_2 = \lim_{s \rightarrow 0} s G_{e_2R}(s) \frac{1}{s} \Leftrightarrow \varepsilon_2 = \frac{1}{1 + K_2 G_{cl,1}(0) G_2(0)} \stackrel{(2)}{=} \frac{1}{1 + K_2(1 - \varepsilon_1) G_2(0)}$$

The DC gain of $G_2(s)$ can be found from the given Bode plot as $G_2(0) = -7.95881$ dB or $G_2(0) = 0.4$. Then from the last relation we have by inserting the numerical values for ε_2 , ε_1 and $G_2(0)$

$$K_2 = \frac{1 - \varepsilon_2}{\varepsilon_2 G_2(0)(1 - \varepsilon_1)} = 79.1667 .$$

Therefore, the only acceptable option is (e).

Question 20

Consider the system described by the following block diagram,

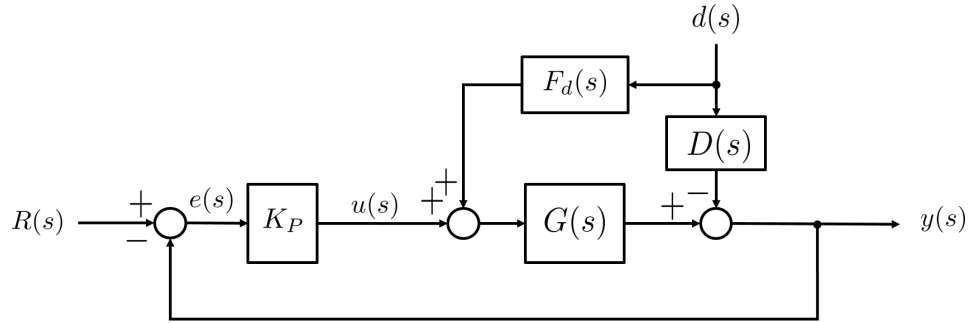


Figure 9: Block diagram.

where $K_P = 1$, $d(s)$ is an external disturbance and the transfer functions $G(s)$ and $D(s)$ are defined as

$$G(s) = \frac{21(0.5s + 1)}{s^2 + 4s + 21}$$

$$D(s) = \frac{1}{0.01s + 1}$$

If $F_d(s)$ is an unfiltered dynamic feed-forward controller designed to compensate for the disturbance, then the sensitivity function related to the disturbance $d(s)$ is:

a) $G_{yd}(s) = 0$. (Correct)

b) $G_{yd}(s) = \frac{100s^2 + 400s + 2100}{s^3 + 114.5s^2 + 1492s + 4200}$. (Used $\frac{D}{1+K_P G}$)

$$\text{c) } G_{yd}(s) = -\frac{100s^2 + 400s + 2100}{s^3 + 114.5s^2 + 1492s + 4200} \cdot \text{(Used } \frac{-D}{1+K_PG} \text{)}$$

$$\text{d) } G_{yd}(s) = \frac{100}{s+1} \cdot \text{(Used } F_dG \text{)}$$

$$\text{e) } G_{yd}(s) = \frac{10.5s^2 + 1071s + 2100}{s^3 + 104s^2 + 1471s + 4200} \cdot \text{(Used } \frac{D}{1+DG} \text{)}$$

Solution: The dynamic feed-forward controller is given by

$$F_d(s) = \frac{D(s)}{G(s)} = \frac{s^2 + 4s + 21}{21(0.5s + 1)(0.01s + 1)}$$

Since the ratio is a stable and proper transfer function, we do not need any filtering. This means that $F_d(s)G(s) = D(s)$ at all frequencies. The sensitivity of the output to the disturbance, i.e. transfer function from $d(s)$ to $y(s)$ is given by

$$G_{yd}(s) = \frac{-D(s) + F_d(s)G(s)}{1 + K_PG(s)} = 0 \text{ .}$$

Therefore, the only acceptable option is (a).